Why 39.9% of signups never see the product work

A two-part analysis on a simulated AI writing assistant: funnel and churn diagnosis on 30,000 users, then an A/B test of a Rewrite-first onboarding.

SQL · Python · Power BI · A/B testing · Statistics

30,000	175,151	+43.7%	0.63	22
SIMULATED USERS	SESSIONS	A/B LIFT (FIRST AI)	CHURN MODEL AUC	SQL QUERIES

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Simulated data — see README

Most users are lost before the product works

39.9%

of signups (11,967) never reach a first AI output

34.9%

of users who DO reach it later churn

36%

of churn lands days 21-27 after first AI output

2.1x

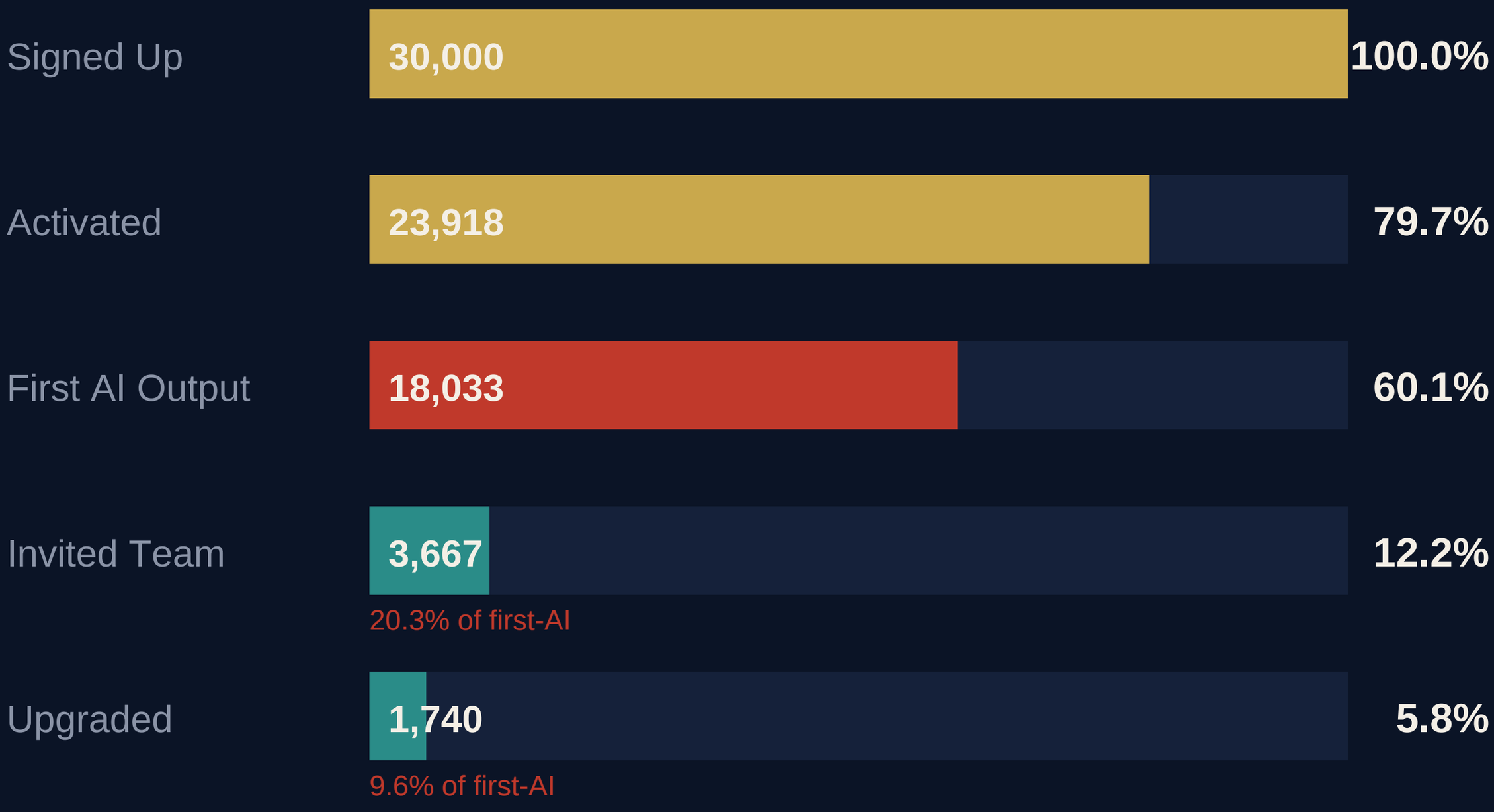
Organic vs Paid Meta week-8 retention

20% / 10%

of first-AI users invite / upgrade

> Users are lost earlier than most funnels assume: before first value, not at the paywall. That is a sequencing problem, so the test targets onboarding.

Where users fall off and which branch is thin



> Invite and upgrade are parallel branches off first AI output, not steps in a sequence, so each is measured against first-AI users.

FIRST AI OUTPUT RATE BY CHANNEL

Organic Search 71.3%	**Referral 68.8%**	**Direct 61.2%**
Paid Google 52.1%	**Paid Meta 41.3%**	

Simulated data — see README

Channel mix decides how long users stay

Channel	Week 1	Week 4	Week 8
Organic Search	98.2%	77.5%	**61.7%**
Referral	97.6%	74.3%	**55.8%**
Direct	98.3%	70.0%	**49.4%**
Paid Google	97.3%	62.1%	**36.5%**
Paid Meta	96.9%	58.3%	**29.3%**

Organic retains users 2.1x better than Paid Meta at week 8 (61.7% vs 29.3%). Retention alone is not a budget decision: it also needs CAC and LTV, which this dataset does not have.

Retained = last session is at least N weeks after signup.

One feature clearly separates stayers

REWRITE

With 28.3% · Without 40.8%

Significant (chi2 = 308.7)

ANALYZE

With 32.4% · Without 37.1%

Significant (chi2 = 43.6)

SUMMARIZE

With 34.5% · Without 35.3%

Not significant (chi2 = 1.1)

GENERATE

With 34.3% · Without 35.4%

Not significant (chi2 = 2.5)

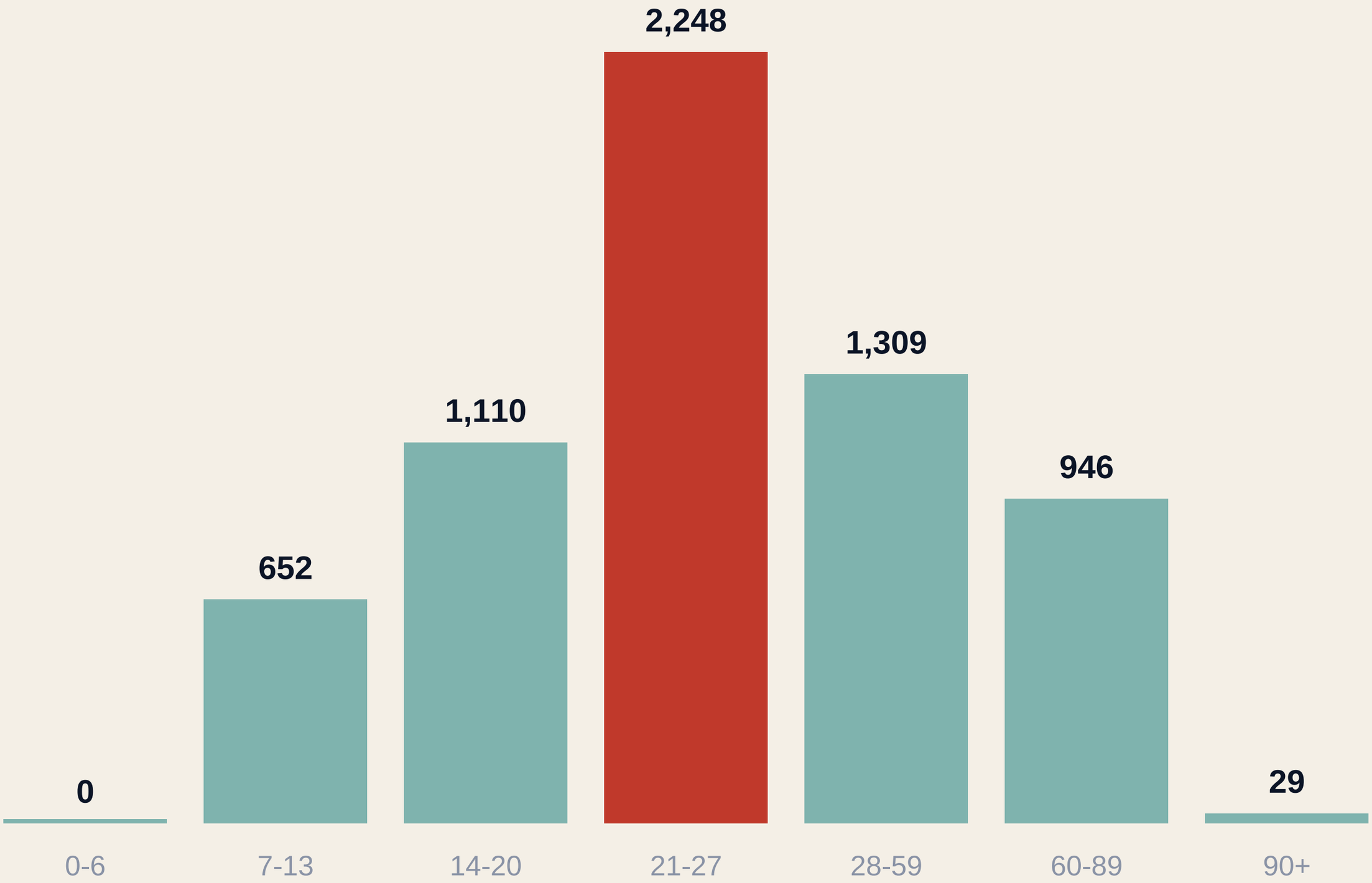
TRANSLATE

With 34.6% · Without 35.1%

Not significant (chi2 = 0.4)

Population: users who reached first AI output. Chi-square, alpha = 0.05.

When users leave after first AI output



Days 21-27 holds 35.7% of all churn. A check-in around day 14 reaches users just before it.

A model that explains churn, but predicts it weakly

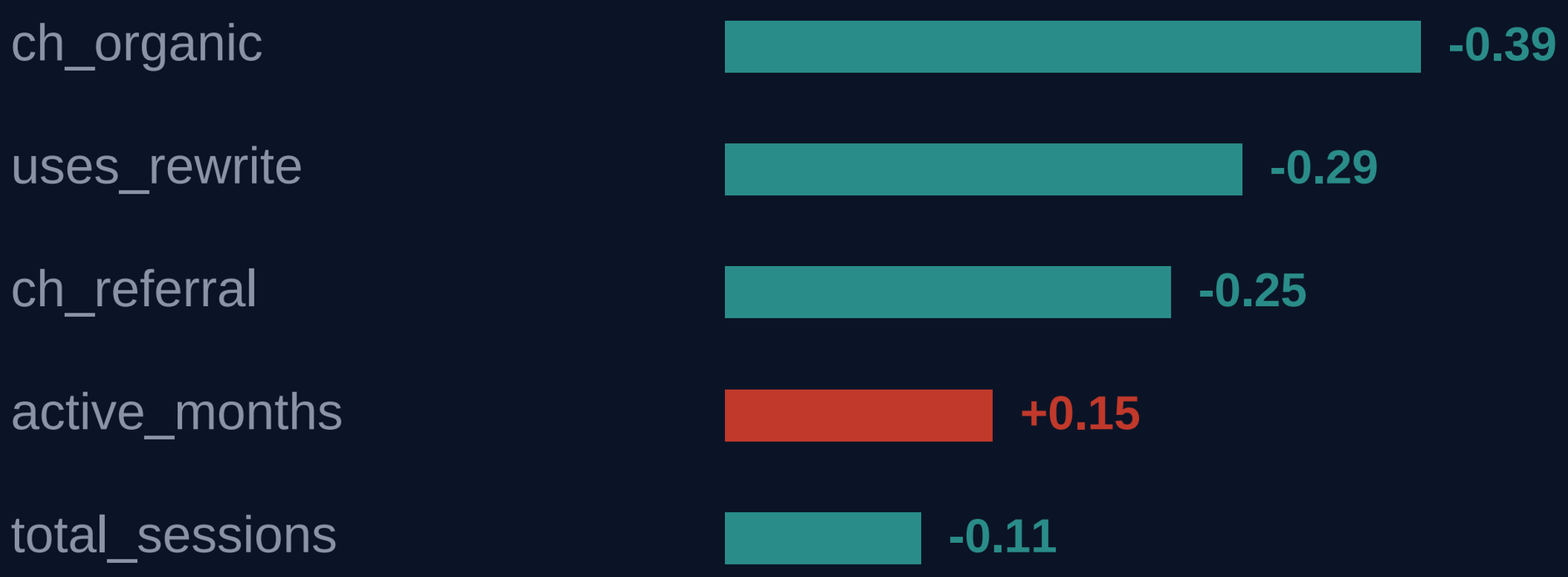
0.63

ROC-AUC, 80/20 split

11

features · 18,033 eligible users

LARGEST COEFFICIENTS (STANDARDISED)



An earlier draft showed AUC 0.75 and 'more features means more churn'. Both came from including users who never reached first AI output (they cannot churn). Removing them fixed it.

Simulated data — see README

Test the fix:
does it move the metric?

6,200 simulated users · 50/50 · Mar–Sep 2025 · alpha 0.05

CONTROL

34.7%

TREATMENT

49.9%

First AI output rate

+43.7%

+15.2 pp · z = 12.1 · p < 0.001

Two-proportion z-test

+65.0%

Week-4 retention

15.8% → 26.1%

+60%

Week-1 sessions / user

0.68 → 1.09

SHIP

Decision

3 of 3 significant

Lift is positive in all five channels (+31% to +67%). With 300–900 users per arm per channel, differences between channels are mostly noise.

Built end to end.
Method over claims.

22 SQL queries · 13 charts · interactive dashboard · 5-page Power BI report · A/B test · churn model.

All data is simulated. The Rewrite effect and the A/B treatment effect are built into the generator, so this shows the workflow, not findings about a real product.

github.com/Ajaypatel06

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